

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M. Tech. in AI & ML

I Semester 2025-2026

Mid-Semester Test
(EC3 - Regular)

Course Number	AIMLCZG511
Course Name	DEEP NEURAL NETWORK
Nature of Exam	Open Book
Weight-age for grading	40
Total Marks	120

# Pages	7
# Questions	7

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1. A hospital is building a diagnostic system to classify medical X-ray images into 10 distinct categories (Normal, Pneumonia, Cardiomegaly, etc.). The system is designed as a Deep Feedforward Neural Network (DFNN) with two hidden layers and a softmax output.
 - (a) A single neuron in Hidden Layer 1 receives the following input from an image patch:
Input: $\mathbf{x} = [0.8, -0.5, 0.2]$, Weights: $\mathbf{w} = [0.4, 0.1, -0.2]$, Bias: $b = 0.5$
 - i. Perform inference through this neuron, given the above details. Assume ReLU activation. (2 marks)
 - ii. The output layer produces 10 raw logit scores. Write the Softmax formula and explain in one sentence why it is appropriate for this 10-class task. (2 marks)
 - iii. Define the activation functions for Hidden Layer 1 (ReLU), Hidden Layer 2 (Leaky ReLU), and the Output Layer (Softmax). Provide the mathematical formula for each. (3 marks)
 - (b)
 - i. Explain why a Perceptron fundamentally cannot solve the 10-class task. Your answer must reference linear separability, decision boundaries, and the XOR problem as an analogy. (4 marks)
 - ii. During training of the DFNN, gradients near the input layers become vanishingly small. Identify the cause, explain the impact on learning, and recommend TWO architectural fixes with justification. (4 marks)

2. A telecom company wants to predict subscriber churn (whether a customer will cancel their subscription) using customer behavioural data: call duration, data usage, number of complaints, and contract type. The team progressively builds from logistic regression to a full DFNN.
- (a)
 - i. Identify the appropriate output activation function and loss function for the binary churn prediction task. Provide the formula for each and justify why they are paired together. (3 marks)
 - ii. The logistic regression model outputs a churn probability of 0.73 for a customer. State the predicted class assuming a threshold of 0.5 and explain what the threshold represents in the churn context. (2 marks)
 - (b)
 - i. The logistic regression model achieves 88% overall accuracy but only 54% recall for churners. Identify TWO specific architectural reasons why a DFNN would better capture the churn signal than logistic regression. (2 marks)
 - ii. The team designs a DFNN with two hidden layers. Justify the choice of ReLU for Hidden Layer 1 and Leaky ReLU for Hidden Layer 2, addressing computational efficiency and the dying ReLU problem respectively. (3 marks)
 - iii. Design the complete DFNN architecture for the churn task. Specify: input dimension, number of hidden layers, neurons per layer, activation functions, loss function, and output configuration. Justify each design choice in the context of the churn prediction problem. (5 marks)

3. A deep learning team at a hospital develops an automated skin cancer detection system using 8,000 dermoscopic images ($512 \times 512 \times 3$) across 7 lesion types. Melanoma has only 600 images while benign nevi has 3,200. A pre-trained ResNet-50 achieves 92% overall accuracy but only 68% sensitivity for melanoma. GPU memory is limited to 8GB with real-time inference $<500\text{ms}$.
- Should the team use transfer learning or train from scratch? Provide THREE specific reasons based on the dataset characteristics. (3 marks)
 - The team replaces the final FC layer of ResNet-50 with: $2048 \rightarrow 512 \rightarrow 7$ classes, with Dropout ($p = 0.5$) between the two FC layers. Calculate: parameters removed from the original FC layer, parameters added by the new classifier including biases, and net change in total parameters. (4 marks)
 - A wildlife conservation team deploys a CNN on solar-powered edge devices ($224 \times 224 \times 3$ input, 10 classes):

Conv(32, 5×5 , $s=1$, $p=2$) $\rightarrow$ MaxPool(2×2 , $s=2$) $\rightarrow$ Conv(64, 3×3 , $s=1$, $p=1$) $\rightarrow$ MaxPool(2×2 , $s=2$) $\rightarrow$ Conv(128, 3×3 , $s=1$, $p=1$) $\rightarrow$ MaxPool(2×2 , $s=2$) $\rightarrow$ Flatten $\rightarrow$ FC($100352 \rightarrow 256$) $\rightarrow$ FC($256 \rightarrow 10$).

 - Calculate total parameters (including biases) in Layer 1 (Conv, 3-channel input) and Layer 8 (first FC layer). (3 marks)
 - The team adds an infrared channel making input $224 \times 224 \times 4$. How many parameters does Layer 1 have now? What is the percentage increase? Do parameters in Layer 3 or Layer 8 change? Explain why or why not. (3 marks)
 - Layer 8 contains 99.2% of all trainable parameters. Three solutions are proposed
Option A: Global Average Pooling: $28 \times 28 \times 128 \rightarrow 128$, then FC($128 \rightarrow 10$);
Option B: Conv(128, 3×3 , $s=2$) reducing to $14 \times 14 \times 128$, then Flatten+FC;
Option C: Conv(32, 1×1) reducing to $28 \times 28 \times 32$, then Flatten+FC.
Choose ONE option, calculate the new parameter count for the modified FC pathway, and state ONE advantage and ONE disadvantage for the edge deployment. (3 marks)
 - The ResNet-50 model achieves 89% validation and 94% training accuracy on full images, but drops to 71% on 128×128 crops from the same images. Explain TWO CNN architectural reasons for this degradation. (2 marks)
 - A colleague claims: “Adding the infrared channel increases the receptive field at Layer 5 because the network has more input information.” Is this correct? Justify with reference to how receptive fields are computed. (2 marks)

4. An industrial edge device performs real-time anomaly detection on sensor streams (vibration, temperature, current) from a factory floor motor. Sensor data arrives continuously at high frequency; anomalies are caused by recent pattern changes; the device has limited memory, compute, and power.

(a) Given input dimension $d = 10$, hidden dimension $h = 20$:

- i. Compute the total trainable parameters (excluding output layer) for a Vanilla RNN and a GRU. (2 marks)
- ii. For the factory edge device scenario above, select the most appropriate model among Vanilla RNN, GRU, and LSTM. Justify your choice addressing memory, compute constraints, and the nature of the anomaly signal. (3 marks)

(b) You are modelling a single time step of the factory sensor signal. All bias terms are zero.

Input: $\mathbf{x}_t = [2, 1]^\top$, Previous hidden state: $\mathbf{h}_{t-1} = [0.5, -0.5]^\top$, $d = 2$, $h = 2$.

- i. For a Vanilla RNN with compute the pre-activation vector and hidden state $\mathbf{h}_t$. (2 marks)

$$W_{hh} = \begin{bmatrix} 0.6 & 0 \\ 0 & 0.6 \end{bmatrix} \quad W_{hx} = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.4 \end{bmatrix}$$

- ii. For a GRU with $W_z = I$, $U_z = 0$, $W_h = 0.5I$, $U_h = 0$, using $\sigma(2) \approx 0.88$, $\sigma(1) \approx 0.73$, $\tanh(1) \approx 0.76$, $\tanh(0.5) \approx 0.46$: compute update gate $\mathbf{z}_t$, candidate state $\tilde{\mathbf{h}}_t$, and final hidden state $\mathbf{h}_t$. (3 marks)
 - iii. Using your numerical results, state ONE numerical reason why the GRU update is more conservative than the Vanilla RNN update. (2 marks)
- (c) Consider a GRU modelling a 3-step sensor sequence. $\mathbf{h}_t = z_t \mathbf{h}_{t-1} + (1 - z_t) \tilde{\mathbf{h}}_t$, with $\partial L / \partial \mathbf{h}_3 = 1$, $z_t = 0.9$ for $t = 2, 3$, candidate path ignored.
- i. Compute $\partial L / \partial \mathbf{h}_1$ by backpropagating from $t = 3$ to $t = 1$. (1 mark)
 - ii. Explain how the GRU update gate mitigates vanishing gradients, and identify ONE scenario where even a GRU struggles with long-range dependencies. (1 mark)
 - iii. Under what specific operational constraints would an RNN remain preferred over a Transformer for the factory anomaly task? Justify with reference to computational complexity. (1 mark)

5. Attention mechanisms are widely used in sequential decision systems. The following sub-questions explore scaled dot-product, additive, and multi-head attention in different application contexts.

- (a) In a product recommendation engine, users browse items sequentially: Phones $\rightarrow$ Bags $\rightarrow$ Headset. Self-attention identifies which past items provide the strongest signal for the next recommendation.

Given embeddings at the third position (Headset):

$$\mathbf{q}_3 = [2, 7], \quad \mathbf{k}_1 = [3, 1], \quad \mathbf{k}_2 = [3, 1], \quad \mathbf{k}_3 = [7, 3], \quad \mathbf{v}_1 = [3, 1], \quad \mathbf{v}_2 = [3, 1], \quad \mathbf{v}_3 = [8, 3]$$

- i. Using scaled dot-product attention, compute the raw scores, apply softmax, and obtain the context output vector for the third position. (3 marks)
 - ii. The deployed model recommends exactly what users bought in training logs but fails to anticipate future purchases in production. Identify the attention mechanism likely missing during training and justify in two sentences. (2 marks)
- (b) Electronic health records contain patient event sequences: Blood Test $\rightarrow$ X-Ray $\rightarrow$ Medication. Additive attention identifies which past events most influence the next clinical decision.

Given embeddings at the third position (Medication):

$$\mathbf{q}_3 = [2, 1], \quad \mathbf{k}_1 = [1, 0], \quad \mathbf{k}_2 = [0, 1], \quad \mathbf{k}_3 = [2, 1], \quad \mathbf{v}_1 = [1, 0], \quad \mathbf{v}_2 = [1, 1], \quad \mathbf{v}_3 = [3, 1]$$

- i. Using additive attention with tanh scoring, compute alignment scores, softmax weights, and the context output vector for the third position. (3 marks)
 - ii. Identify the most appropriate attention type for each scenario and justify in no more than two sentences:
Scenario 1: A clinical system combines patient records with external drug interaction databases.
Scenario 2: A patient record contains diverse signals (lab results, imaging, vitals, prescriptions) where both consecutive imaging and medication-lab correlations are diagnostically important. (2 marks)
 - iii. Compare scaled dot-product attention and additive attention in terms of computational complexity and expressiveness. Under what conditions would you prefer one over the other? (2 marks)
- (c) A medical AI team debates between single-head and multi-head attention for their clinical NLP system.
- i. Given $d_{model} = 512$ and $h = 8$ heads, compute the dimension of each head's Q/K/V projections and compare total parameter counts for single-head vs. 8-head attention. (1 mark)
 - ii. The single-head model attends only to the most recent clinical event, ignoring older but critical diagnostic signals. Explain how multi-head attention resolves this, describing what different heads learn in the clinical context. (1 mark)
 - iii. Identify ONE limitation of multi-head attention in a resource-constrained clinical deployment and suggest a mitigation. (1 mark)

6. A healthcare NLP team is building a clinical report summarisation system that processes long patient discharge notes and generates concise summaries. They are evaluating transformer architectures for this task.
- (a)
 - i. The team removes positional encoding to simplify the architecture. Explain what positional encoding does and describe the specific failure mode that will be observed in clinical text without it. (3 marks)
 - ii. The team is choosing between encoder-only, decoder-only, and full encoder-decoder transformer. Match each architecture to ONE of the following tasks and justify: (a) clinical named entity recognition, (b) discharge note generation, (c) medical report summarisation. (4 marks)
 - (b) The encoder processes a tokenised clinical sentence with $d_{model} = 512$, 8 attention heads, $d_{ff} = 2048$, sequence length $L = 128$.
 - i. Trace the tensor dimensions at each stage of one encoder layer: input $\rightarrow$ multi-head attention output $\rightarrow$ add & norm $\rightarrow$ feed-forward output $\rightarrow$ add & norm. (3 marks)
 - ii. Compute total trainable parameters in ONE encoder layer: multi-head attention (Q, K, V projections + output projection) and feed-forward sublayer. (3 marks)
 - iii. Training is unstable in early epochs. A colleague suggests Pre-LN (normalisation before sublayer) rather than Post-LN (after sublayer). Compare both in terms of training stability and final performance. (3 marks)
 - (c) The team must choose between fine-tuning BERT-style, GPT-style, or training T5-style from scratch for clinical summarisation. They have 50,000 annotated note-summary pairs and 4 A100 GPUs.
 - i. Compare the three architectures for this task: state the pre-training objective, architectural form, and suitability for summarisation for each. (2 marks)
 - ii. Self-attention has $O(L^2)$ complexity. Clinical notes can be up to 2,048 tokens. Explain why this is a bottleneck and propose ONE mitigation strategy with its trade-off. (2 marks)

7. A team is training a deep neural network for financial fraud detection. The following sub-questions are based on training curves and regularisation behaviour observed during experimentation.

- (a) Training loss curves for four optimizers (SGD, SGD+Momentum, RMSprop, Adam) over 100 epochs show:

Curve A: Smooth but very slow descent, still decreasing at epoch 100.

Curve B: Fast initial descent with oscillations, settles near a good minimum.

Curve C: Fast descent, minimal oscillation, plateaus early at a higher loss than D.

Curve D: Fastest convergence, smooth descent, reaches lowest loss.

Match each curve to its optimizer and justify each match in one sentence. (4 marks)

- (b) i. Three learning rate schedules applied to the same Adam model produce the following validation accuracy curves:

Schedule X: Accuracy rises sharply, then drops and oscillates.

Schedule Y: Accuracy rises steadily, plateaus at epoch 40, no further improvement.

Schedule Z: Accuracy rises steadily, brief dip at epoch 30, recovers to highest final accuracy.

Identify the likely schedule type for X, Y, Z from: constant high LR, constant low LR, cosine annealing with warm restarts. Justify each. (4 marks)

- ii. After epoch 60, validation loss rises while training loss continues falling under Adam. Identify this phenomenon and suggest ONE optimizer-level intervention (not architectural) to address it. (3 marks)

- iii. The fraud detection model shows the following across four regularisation runs:

Run 1: Train 98%, Val 71% — large gap persists.

Run 2: Train 84%, Val 83% — small gap, both plateau early at suboptimal values.

Run 3: Train 91%, Val 90% — small gap, smooth convergence.

Run 4: Train 76%, Val 74% — noisy, slow, unstable.

Diagnose the regularisation condition for each run and justify. (4 marks)

- iv. For Run 1, propose a combined regularisation strategy using TWO techniques. For each, explain the mechanism by which it reduces the train-validation gap. (3 marks)

- (c) The team adds Batch Normalisation to the network in Run 4. Predict the effect on training stability and convergence speed, and explain the mechanism behind your prediction. (2 marks)